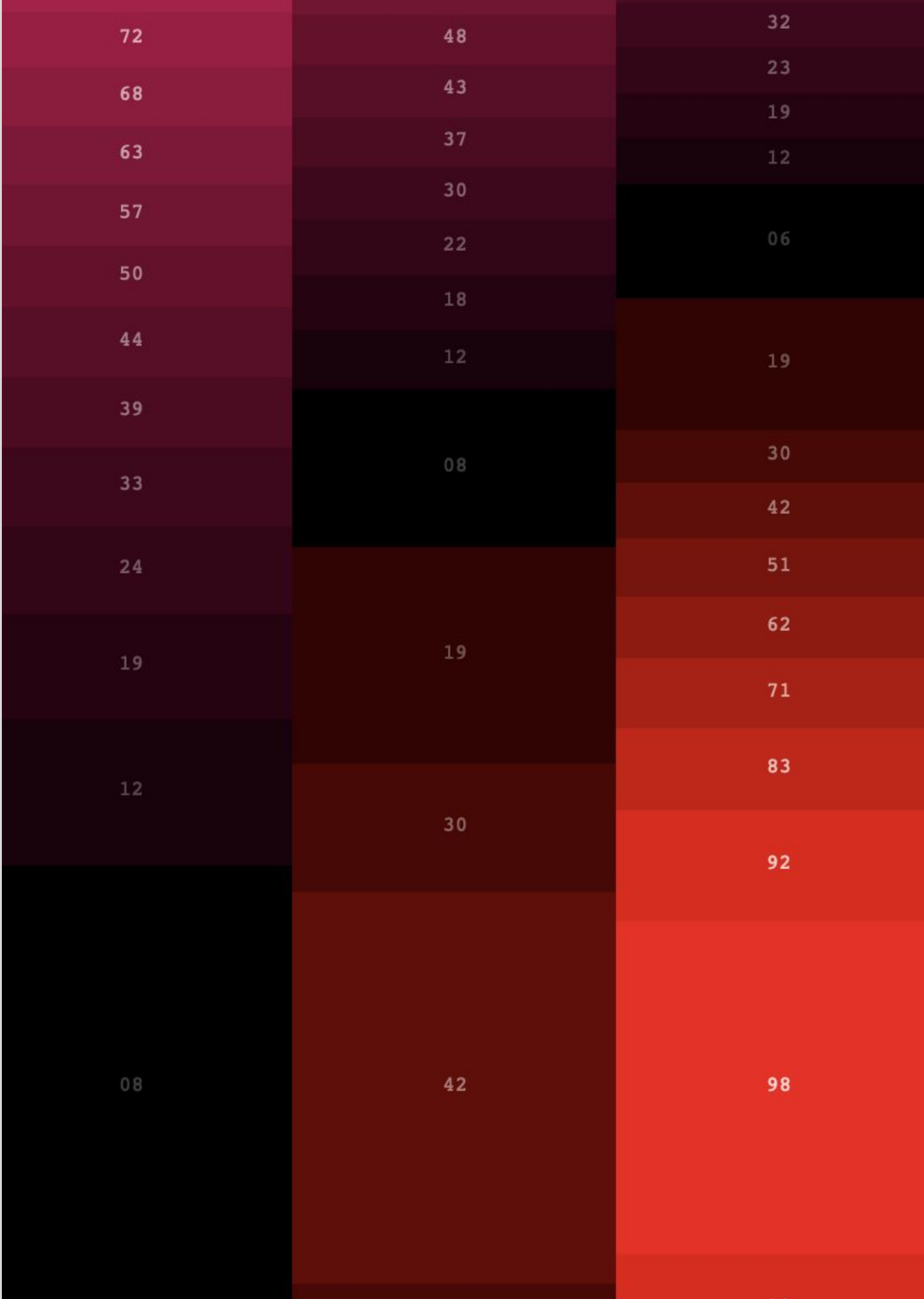


CURTIS GALIONE
SOLUTIONS ENGINEERING
BRAINTRUST

EVALS FOR ENGINEERS



OUTLINE

- Introduction
- What and why eval?
- How eval?
- Do eval
- Wrap up

There's no free lunch

- ✓ End-to-end workflow automation → 😬 Error propagation without human oversight
- ✓ Scale knowledge work → 😬 Overreliance, hallucinations, and skill atrophy
- ✓ Personalize customer experiences → 😬 Privacy, compliance, and brand risk
- ✓ New revenue models → 😬 Regulatory uncertainty and security vulnerabilities
- ✓ Outsourcing massive computations → 😬 Potential for runaway cost

Evals are simply a way to systematically measure and improve an AI application



Grow your coverage
Iterate faster
Ship with confidence

Evals are not unit tests

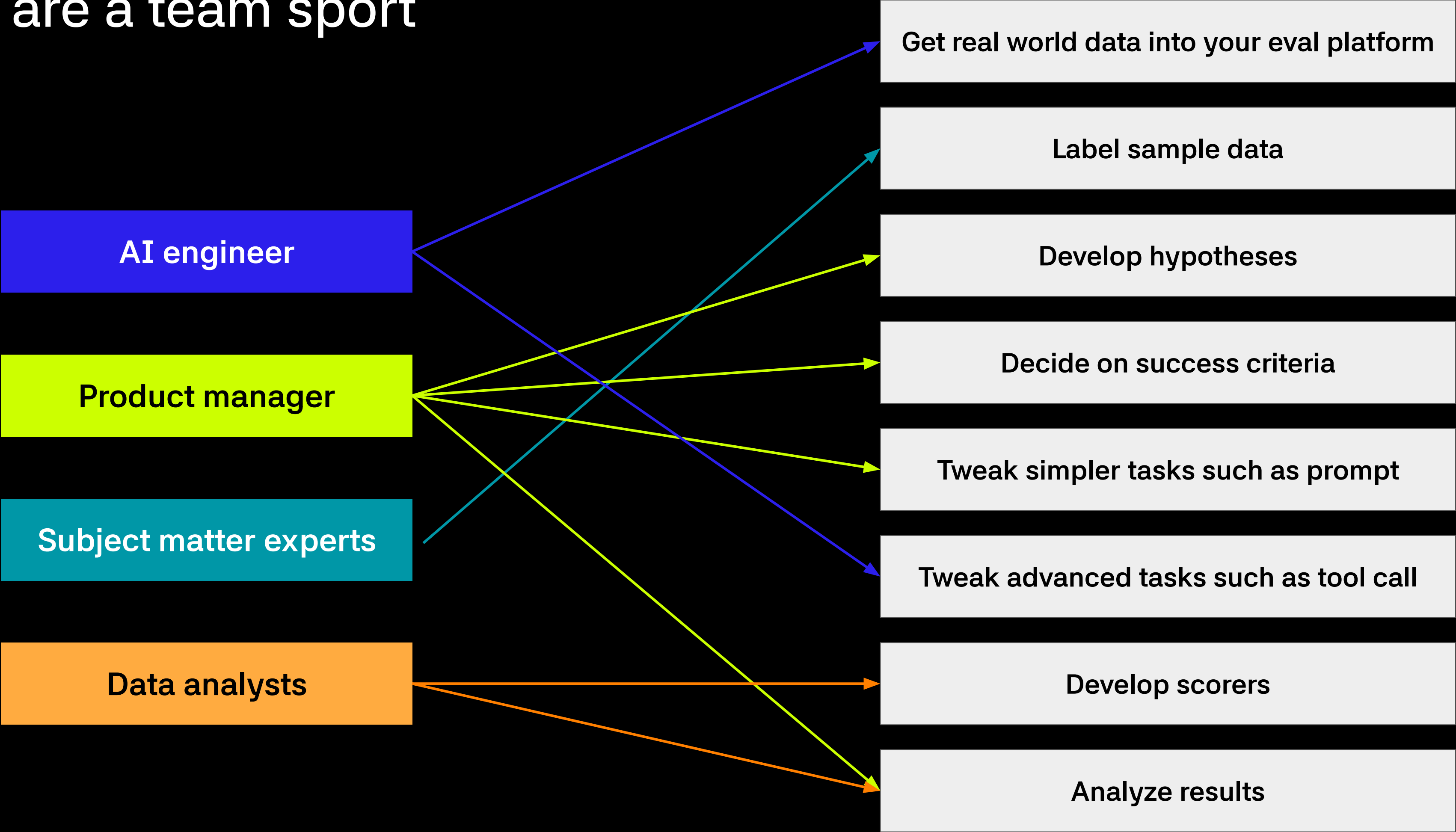
Traditional testing

- Deterministic problems = deterministic solutions
- Be exhaustive and test all possible inputs
- The more unit tests the better
- Testing "does it work?" and "do I have bugs?"

Evals

- We ❤️ LLMs because of their variety
- No possible way to test all inputs/outcomes
- Built around failure modes
- Need to work within Pareto principles
- Rerunning production; performance, not just errors
- "Am I avoiding failure modes?"

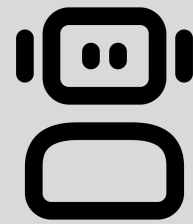
Evals are a team sport



How much grounding info do you need to score this failure mode?

Behavior	Context	Ground truth*
Judging the overall behavior of the agent.	Judging how well the agent is gathering and using context from other systems	Judging the agent's performance based on expert information
Knowledge available: requirements for you want the agent to behave	Knowledge available: the context the agent gained along the way	Knowledge available: Expert annotated information
*cannot be done "online"		
Amount of grounding information		

Scoring techniques



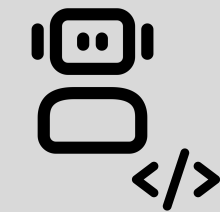
LLM-as-judge

A prompt and LLM creates a score based on information in the agent trace. Useful for subjective measurement.



Code-based

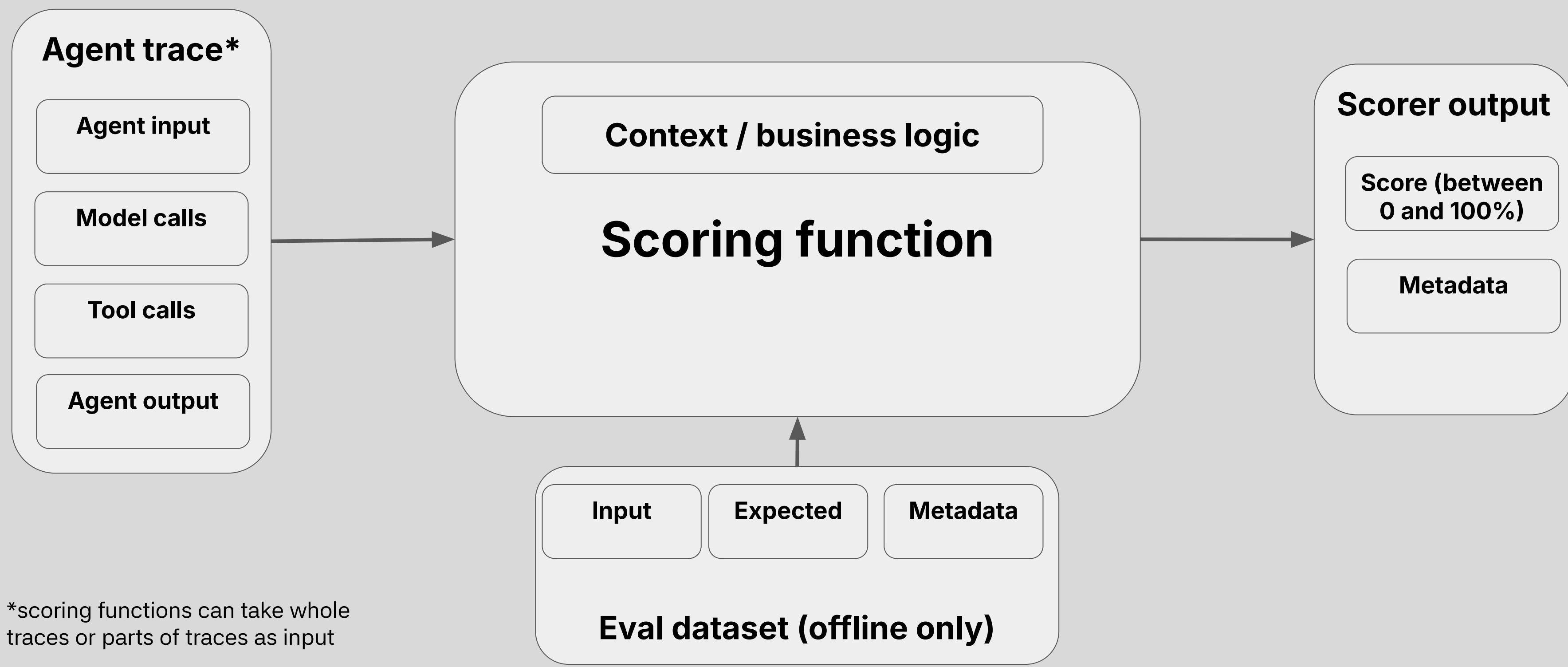
Extract data from the agent trace and run arbitrary code on it. Highly customizable and good for deterministic scoring.



Judge + code

Why not both? Mix in both subjective and deterministic logic. Very flexible. Works best when using structured outputs from the LLM.

Components of scoring functions



*scoring functions can take whole traces or parts of traces as input

Exercise!

git clone <https://github.com/braintrustdata/evals-for-engineers.git>

